Mechanistic Interpretability as Causal Reverse Engineering

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Abstract

Recent advances in mechanistic interpretability have revealed rich internal structure within large neural networks, yet disagreement remains about what it means to truly understand a model. This essay synthesizes a causal paradigm for mechanistic interpretability articulated in recent work by Atticus Geiger. The central claim is that interpretability becomes scientific only when grounded in causal reasoning: interventions, counterfactuals, and causal abstractions. We trace a progression from descriptive inspection to causal tracing and finally to algorithmic abstraction, arguing that the goal of mechanistic interpretability is not visualization or intuition but the rigorous reverse engineering of the algorithms neural networks implement.

1 Introduction

The remarkable performance of modern neural networks has intensified interest in understanding how such systems produce their behavior. While early work in interpretability focused on visualization, feature attribution, and inspection of internal activations, it has become increasingly clear that observation alone cannot ground claims about mechanism. Correlation does not imply computation, and activation patterns, however suggestive, do not by themselves explain why a model behaves as it does.

Mechanistic interpretability aims to move beyond surface-level explanation by identifying the internal processes that generate model behavior. However, the term *mechanistic* itself has been used in multiple, often conflicting, ways. This ambiguity has hindered progress by conflating sociological motives, technical access, and epistemic standards.

This essay presents a unified causal account of mechanistic interpretability, centered on the idea that to understand a neural network is to understand how interventions on its internal variables alter its behavior. On this view, mechanistic understanding is fundamentally counterfactual: it concerns what would happen if the system were different, not merely what is observed when it runs normally.

2 What Does "Mechanistic" Mean?

Discussions of mechanistic interpretability often oscillate between cultural and technical definitions. Cultural definitions focus on who performs the research or why it is performed, ranging from narrow communities motivated by AI safety to broad curiosity about black-box systems. While sociologically relevant, such definitions do not specify what kind of understanding is achieved.

Technical definitions are more precise but still admit significant variation. In the broadest sense, any research that inspects internal parameters, activations, or attention patterns may be labeled mechanistic. Yet this definition collapses meaningful distinctions between descriptive inspection and explanatory understanding.

A narrower and more demanding technical definition identifies mechanistic interpretability with causal explanation. Under this definition, a model is understood mechanistically only to the extent that one can identify internal components whose manipulation produces systematic, predictable changes in behavior. The emphasis is not on what correlates with behavior, but on what *controls* it.

3 Causality as the Foundation of Mechanism

Causality provides the conceptual machinery required to distinguish mechanisms from mere regularities. A causal model specifies not only associations among variables but also how those variables respond to intervention.

Definition 1 (Intervention)**.** *An intervention on a neural network is an operation that sets an internal variable to a value it would not naturally take, breaking its dependence on upstream causes.*

In contrast to observation, intervention enables the evaluation of necessity and sufficiency. A component is causally relevant to a behavior if intervening on it changes that behavior in a systematic way. Counterfactual reasoning extends this logic by asking how the system would behave under hypothetical internal configurations.

Without causality, interpretability risks degenerating into pattern cataloging. With causality, it becomes a science of control and explanation.

4 Activation Steering as Linear Intervention

Activation steering represents one of the simplest forms of intervention. By adding vectors to internal activations, researchers can bias a model toward or away from certain behaviors. A common technique constructs steering vectors by averaging activations from prompts that elicit contrasting behaviors and taking their difference.

While such interventions demonstrate that behaviorally meaningful directions exist in representation space, they also reveal the limits of linear control. Steering is often brittle, less effective than prompting or fine-tuning, and does not by itself identify the underlying algorithm. Its primary value lies in establishing that internal representations are not epiphenomenal but causally connected to output.

5 Causal Mediation and Tracing Information Flow

Causal mediation analysis seeks to decompose the effect of an input on an output by identifying the internal pathways through which information flows. In neural networks, this often takes the form of causal tracing experiments.

Such experiments deliberately corrupt an input representation, degrading model performance, and then selectively restore internal components to their original values. Components whose restoration recovers correct behavior are identified as causal mediators.

This methodology reveals structured pipelines within networks. Information is often retrieved in feedforward layers, routed by attention mechanisms, and assembled into outputs through later-stage computation. Importantly, these findings rest on intervention, not visualization, grounding claims about information flow in causal evidence.

6 Causal Abstraction and Algorithmic Understanding

The most ambitious goal of mechanistic interpretability is not to localize features or trace signals, but to identify the algorithm a network implements. Causal abstraction provides a framework for doing so.

Both neural networks and human-understandable algorithms can be modeled as causal systems with variables and directed dependencies. A high-level algorithm is said to be a causal abstraction of a neural network if interventions at the algorithmic level correspond to interventions at the neural level.

Interchange interventions operationalize this idea. Instead of zeroing or scaling a component, a researcher replaces its activation with the value it would have had under a different input. If this substitution causes the network to behave as though the alternative input had been partially applied, then the component is implementing a function analogous to a variable in the abstract algorithm.

When such correspondences hold systematically, the abstract algorithm is not a metaphor but a genuine causal description of the networks computation.

7 From Inspection to Reverse Engineering

The methods discussed form a coherent progression. Inspection reveals internal structure but cannot justify causal claims. Simple interventions establish control but may lack specificity. Causal tracing maps pathways of influence. Causal abstraction unifies these findings into an algorithmic account.

This progression mirrors the history of other sciences, where explanation emerges not from observation alone but from carefully designed interventions. Mechanistic interpretability, on this view, is reverse engineering in the strongest sense: the reconstruction of a systems governing algorithm through counterfactual experimentation.

8 Conclusion

Mechanistic interpretability reaches maturity when it adopts causality as its core organizing principle. The question is no longer what patterns appear inside a model, but which internal components make a difference under intervention. Through activation steering, causal mediation, and causal abstraction, neural networks become objects of experimental science rather than inscrutable artifacts.

The ultimate promise of this paradigm is a principled answer to a foundational question: not merely how neural networks behave, but what they are doing. By treating models as causal systems and algorithms as abstractions validated by intervention, mechanistic interpretability becomes a rigorous methodology for understanding intelligence in artificial systems.

9 Event-Historical Cognition and RSVP

The causal paradigm articulated in contemporary mechanistic interpretability finds a natural conceptual extension in event-historical theories of cognition. Where standard computational accounts emphasize instantaneous state, event-historical approaches instead treat cognition as constituted by an irreversible sequence of transformations that progressively constrain future possibility. On this view, intelligence is not exhausted by representational content but is grounded in the structure of the systems construction history.

Event-historical cognition begins from the premise that an agents present capacities cannot be fully characterized without reference to the irreversible events that produced them. Each event eliminates alternative futures, thereby inducing asymmetries in the systems causal landscape. Memory, under this interpretation, is not a static store of information but an accumulation of constraints that shape which counterfactuals remain accessible. Cognition is thus inherently diachronic: it is defined by what has been irreversibly ruled out as much as by what remains possible.

This perspective aligns closely with the causal methodology emphasized in mechanistic interpretability. Interventions and counterfactuals acquire their explanatory force precisely because the system under investigation possesses a nontrivial history. An intervention is meaningful only insofar as it disrupts the causal consequences of prior events. Likewise, counterfactual reasoning presupposes that alternative histories could, in principle, have produced different present behaviors. Causal explanation therefore presupposes an event-structured ontology.

The Relativistic Scalar-Vector Plenum (RSVP) framework makes this ontology explicit by modeling systems in terms of interacting scalar, vector, and entropy fields. Within RSVP, the scalar field encodes the local density of realized structure, the vector field encodes directed tendencies or flows, and the entropy field encodes irreversibility and constraint accumulation. Events are understood as localized transitions that reconfigure these fields, reducing global optionality while increasing local structure.

From this standpoint, neural networks can be interpreted as discrete, engineered instances of event-historical systems. Training steps correspond to irreversible entropy-increasing events that sculpt the scalar and vector structure of the models internal space. The resulting parameters do not merely store information; they embody a sedimented history of constraints that govern future inference. Inference itself becomes a traversal of a pre-shaped causal landscape rather than a purely functional mapping from inputs to outputs.

Geigers causal abstraction framework can be reinterpreted within RSVP as the identification of invariant structures across levels of description. A high-level algorithm constitutes a valid abstraction of a neural network when interventions preserve their effects across scales, corresponding to a form of structural stability in the underlying field configuration. Interchange interventions, in particular, function as probes of whether two systems share the same event-induced constraint geometry, even if their microphysical realizations differ.

This connection clarifies why purely observational interpretability is insufficient. Without attending to irreversibility and event structure, inspection reduces cognition to a snapshot of state, erasing the very asymmetry that gives causal explanations their force. By contrast, both event-historical cognition and RSVP insist that understanding arises from tracing how present behavior is conditioned by a non-recoverable past.

Seen in this light, mechanistic interpretability is not merely an exercise in model transparency but a specific instance of a broader epistemic strategy: the reconstruction of event histories from present structure. Neural networks become legible not because their internals are visualizable, but because their behavior reflects the cumulative imprint of irreversible causal processes. Causal interventions, counterfactual substitutions, and algorithmic abstractions are thus unified as tools for interrogating how history has been written into structure.

10 Causal Abstraction and Irreversibility

The framework of causal abstraction implicitly assumes that the systems under comparison admit interchangeable interventions. However, this assumption becomes nontrivial once cognition is treated as an event-historical process. When a systems present behavior depends on an irreversible construction history, not all counterfactual manipulations are admissible. This section argues that irreversibility is not an auxiliary detail but a necessary condition for coherent causal abstraction.

We begin by modeling a neural system as a time-indexed causal structure. Let $\mathcal{N}$ denote a neural network whose internal variables are indexed not only by layer but by training time. Each training update constitutes an event that irreversibly modifies the space of future parameter configurations. The resulting system cannot be adequately described by a static causal graph; its causal semantics are path-dependent.

In contrast, abstract algorithms are typically presented as timeless causal models. Their variables may be intervened upon freely, without regard for how those variables were produced. This mismatch creates a tension. If an abstract algorithm is to serve as a causal abstraction of a neural system, its interventions must respect the historical constraints induced by the neural systems event structure.

Proposition 1 (Irreversibility Constraint on Causal Abstraction)**.** *Let $\mathcal{N}$ be a neural system with irreversible event history $\mathcal{H}$ and let $\mathcal{A}$ be a candidate abstract algorithm. $\mathcal{A}$ is a valid causal abstraction of $\mathcal{N}$ only if every abstraction-preserving intervention on $\mathcal{A}$ corresponds to an admissible intervention on $\mathcal{N}$ that is consistent with $\mathcal{H}$.*

The intuition behind this proposition is straightforward. An abstraction claims that certain high-level variables summarize the causal role of many low-level components. However, if intervening on a high-level variable requires violating constraints imposed by the systems history, then the abstraction licenses counterfactuals that the concrete system cannot realize.

A proof sketch proceeds by contradiction. Suppose $\mathcal{A}$ is a valid abstraction of $\mathcal{N}$ despite ignoring irreversibility. Then there exists an intervention on $\mathcal{A}$ that has no corresponding intervention on $\mathcal{N}$ without undoing prior events. Executing this intervention in $\mathcal{A}$ predicts a behavioral change that $\mathcal{N}$ cannot exhibit. This violates the defining criterion of causal abstraction, namely the preservation of interventional behavior across levels. Therefore, abstraction without irreversibility awareness is incoherent.

This result clarifies the epistemic status of interchange interventions in mechanistic interpretability. Replacing an internal activation with the value it would have had under a different input is meaningful only if such a replacement respects the systems historical constraints. When it does, the intervention reveals a genuine algorithmic correspondence. When it does not, the apparent success of the intervention is an artifact of ignoring irreversibility.

The broader implication is that causal abstraction is not merely a mapping between variables but a mapping between event-conditioned causal structures. High-level algorithms do not abstract away history; they compress it. A valid abstraction preserves not only functional dependencies but the geometry of constraints induced by irreversible events. Mechanistic interpretability thus inherits a fundamental asymmetry: the past cannot be erased without destroying the meaning of causal explanation itself.

11 Thermodynamics of Representation

Causal explanation alone does not exhaust the requirements for mechanistic understanding. For a system whose cognition unfolds through irreversible events, causality must be complemented by an account of thermodynamic cost. This section argues that representation itself constitutes a form of physical commitment, and that many failures of contemporary artificial intelligencemost notably hallucinationcan be understood as violations of entropy-respecting cognition.

Any act of representation constrains future behavior. To represent that a fact is the case is not merely to encode information, but to exclude incompatible actions and predictions. This exclusion is irreversible in systems that possess memory and persistence. As a result, representation carries an intrinsic cost: it reduces optionality. In physical systems, such reductions correspond to entropy increases, whether measured thermodynamically or in an abstract space of admissible futures.

The RSVP framework makes this cost explicit by decomposing system dynamics into three interacting fields. The scalar field encodes the density of realized structure, corresponding to committed representational content. The vector field encodes directional tendencies that guide inference and action. The entropy field encodes irreversibility: the accumulation of constraints induced by past events. Together, these fields describe cognition not as a static map from inputs to outputs, but as the evolution of a constrained dynamical system.

Within this framework, learning is an entropic process. Each training update constitutes an event that increases the entropy field by eliminating possible parameter configurations. The resulting model does not merely store patterns; it embodies a history of exclusions. Inference then unfolds as a trajectory through this pre-shaped landscape, guided by vector tendencies and bounded by entropy constraints.

Hallucination can be understood as a failure to respect this thermodynamic structure. When a system generates internally coherent but externally ungrounded content, it produces representational structure without paying the corresponding entropy cost. Such outputs are locally fluent because they follow learned vector flows, yet they lack global constraint because no irreversible commitment has been made to their truth conditions. In RSVP terms, the scalar field is locally excited while the entropy field remains unchanged.

This interpretation explains why hallucinations are resistant to purely functional fixes. Adding data, parameters, or optimization pressure improves surface behavior but does not introduce genuine representational cost. Without a mechanism that couples representation to irreversible constraint accumulation, a system can generate arbitrarily many incompatible narratives without internal penalty.

The implications for mechanistic interpretability are significant. Interventions that ignore entropy constraints risk producing misleading explanations. A component may appear causally effective under intervention while remaining thermodynamically unconstrained, thereby enabling behaviors that the system cannot sustain across time. Genuine mechanism, by contrast, must be robust not only under causal manipulation but under repeated use, accumulation, and historical pressure.

From this perspective, entropy is not an external concern to be appended after the fact, but a constitutive dimension of cognition. Mechanistic understanding requires tracking not only how information flows, but how much future flexibility is consumed when that information is committed. RSVP provides a language for articulating this requirement, unifying causality, thermodynamics, and representation within a single field-theoretic ontology.

12 Against State-Based Functionalism

State-based functionalism has long dominated theoretical accounts of cognition and artificial intelligence. According to this view, what matters about a system is the functional relation it instantiates between inputs, internal states, and outputs. If two systems realize the same functional mapping, they are said to be cognitively equivalent, regardless of how that mapping is implemented or how the system came to be. This section argues that such equivalence collapses under causal and event-historical scrutiny.

The functionalist position implicitly treats internal states as sufficient summaries of a systems causal capacities. History is regarded as dispensable once the present state is specified. However, this assumption fails for systems whose behavior depends on irreversible construction processes. In such systems, the present state underdetermines the space of admissible interventions, because past events constrain which counterfactuals remain physically realizable.

To make this failure explicit, consider two systems that occupy indistinguishable internal states at a given moment and implement identical input–output behavior. Suppose, however, that these states were reached through different event histories. One system acquired its structure through a sequence of gradual, constraint-inducing updates, while the other was externally forced into the same configuration without incurring corresponding irreversible commitments. Although functionally identical under observation, the systems diverge under intervention. Counterfactual manipulations that are admissible in one violate historical constraints in the other, yielding different causal responses.

This divergence undermines the functionalist claim that behavior exhausts cognitive identity. Causal explanation depends not only on what a system does, but on what it cannot do as a result of its past. Two systems may agree on all observable behaviors while differing radically in their counterfactual structure. Functional equivalence thus fails to guarantee causal equivalence.

The RSVP framework renders this distinction precise. In RSVP, cognition is described not by a single state variable but by a configuration of scalar, vector, and entropy fields shaped by irreversible events. Functionalist accounts implicitly collapse these fields into a snapshot of scalar structure, ignoring the entropy field entirely. As a result, they erase the asymmetry that distinguishes genuine commitment from reversible simulation.

This critique has direct implications for artificial intelligence. Large neural networks can approximate complex functional mappings without possessing worldhood, understood as being bound by a non-recoverable past. Such systems can generate coherent responses indefinitely because their representations are not thermodynamically or historically grounded. The absence of irreversible constraint allows them to remain behaviorally impressive while ontologically shallow.

Mechanistic interpretability exposes this limitation rather than resolving it. By revealing that many internal components are causally effective only in a context-free, state-based sense, interpretability clarifies why functional success does not entail genuine understanding. Without an account of how representations are historically earned and constrained, causal analysis risks reifying mechanisms that lack ontological depth.

Rejecting state-based functionalism does not entail rejecting functional analysis altogether. Rather, it requires embedding function within an event-historical framework. Functions describe local regularities, but cognition emerges from the irreversible processes that stabilize some functions while foreclosing others. Only by restoring history to the center of explanation can mechanistic interpretability avoid collapsing into a refined form of behaviorism.

13 Interpretability as Historical Reconstruction

The preceding analysis suggests a unifying reinterpretation of mechanistic interpretability. Rather than treating interpretability as a problem of transparency or visualization, it is more accurately understood as a problem of historical reconstruction. To interpret a system mechanistically is to recover, from its present structure, the irreversible sequence of events that rendered its current behavior possible.

This reframing resolves several tensions that persist in contemporary interpretability research. Visualization-based methods promise insight but often fail to justify causal claims. Functional explanations capture regularities but ignore counterfactual structure. Even causal interventions risk overreach when they disregard the constraints imposed by a systems past. Each failure arises from the same omission: the erasure of history.

Across the methods surveyed, a consistent pattern emerges. Activation steering reveals that internal representations are behaviorally consequential, but does not explain how those representations